

Diana (Jiamei) Wang

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EDUCATION

Massachusetts Institute of Technology (MIT)

Expected May 2027

- Candidate for Bachelor of Science in Electrical Engineering & Computer Science (GPA: 4.9/5.0)
- **Relevant Coursework:** Power Electronics Lab, Semiconductor Electronics & Circuits, Analog Electronics Lab, Signal Processing, Solid State Circuits, Digital Systems Laboratory, Dynamic Systems Modeling & Control Design, Electromagnetic Fields & Forces

EXPERIENCE

Schneider Electric

Andover, MA

Power Electronics Engineer Intern

June 2026 – August 2026

- Designed and simulated a 100 kW, 800 V-to-52 V three-phase LLC converter for AI data-center power delivery; selected SiC/GaN devices and specified resonant tank and transformer requirements.
- Built a Python automation framework for PLECS simulations of electrical-stress, loss, and frequency-response at different load conditions, establishing a streamlined workflow with device selection and transformer & controller design.
- Configured hybrid hysteretic charge (HHC) control and developed thermal/loss models in PLECS, predicting 97% nominal-load system efficiency; designed the secondary-side FET board schematic.

MIT Power Electronics Research Group

Cambridge, MA

Undergraduate Researcher

January 2025 – Present

- Designed and experimentally evaluated a 30 MHz GaN Class-E power amplifier; measured 350 W output at 84% efficiency and demonstrated output power modulation over 33–250 W.
- Coauthored a COMPEL 2026 paper as second author, contributing to one of the first demonstrations of high-efficiency switched-mode power conversion with wide-range power control at 30 MHz and hundreds of watts.
- Optimized tank parameters and GaN device package selection in LTspice to maximize efficiency within cooling constraints; designed the PCB layout to reduce high-frequency parasitics; tuned resonant components for soft-switching using impedance measurements and device waveforms.
- Optimized an existing magnetic-cored inductor design using ANSYS Maxwell & Icepak simulations, reducing volume by 50% while retaining $Q > 1000$ at 13.56 MHz with $< 20\%$ degradation from the baseline. Second author on a paper published in the [IEEE Open Journal of Power Electronics](#).

Circuits and Electronics Course

Cambridge, MA

Lab Assistant

September – December 2025, September 2026 - Present

- Providing circuit debugging support in a 200+ student course through projects such as bass-boost filters and light-following robots. Guiding students in linear circuit modeling and analysis techniques during weekly office hours and labs.

PROJECTS

Power Electronics Lab Course

Cambridge, MA

Course Projects

September 2025 – December 2025

- Designed and tested a 12 V boost supply for Class-D amplifier gate drivers and a coupled-inductor SEPIC supply for a DC motor in a Leslie speaker build; verified the amplifier's Butterworth filter frequency response.
- Designed and tested flyback converter, DCM boost LED-driver, resonant ballast, and three-phase inverter circuits.

CMOS Differential Amplifier Tapeout

Cambridge, MA

Integrated Circuit Design Project

March 2026 – May 2026

- Designed a CMOS differential amplifier in Cadence; schematic simulations yielded > 30 dB gain at 1 MHz with < 200 μ W power consumption and > 300 mV peak-to-peak output swing.
- Completed DRC/LVS-clean layout and submitted the design for fabrication; electrical test pending.

Optimized RISC-V Processor

Cambridge, MA

Digital Design Project

February 2025 – May 2025

- Built, verified, and debugged a five-stage RISC-V processor in simulation; reduced MNIST inference runtime by 80% versus the unoptimized pipeline through custom instructions and cache optimizations.

SKILLS

Programming: Python, C, MATLAB, Verilog/SystemVerilog, RISC-V Assembly, Git, Bash

Design & Simulation: Altium, KiCad, Cadence, LTspice, PLECS, ANSYS Maxwell/Icepak

Laboratory: Oscilloscopes, multimeters, impedance analyzer, logic analyzers, soldering